ALICE L WALTON SCHOOL OF MEDICINE

Small Intestine and Colon Disorders

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Learning Objectives (4)

After completing this brick, you will be able to:

1. Differentiate small intestine disorders by presentation, pathophysiology, diagnosis, and treatment.
2. Analyze the presentation, pathophysiology, diagnosis, and treatment of appendicitis.
3. Differentiate colon disorders by presentation, pathophysiology, diagnosis, a
4. Differentiate gastrointestinal obstructions by presentation, pathophysiology, diagnosis, and treatment.

CASE CONNECTION

A 22-year-old woman with Anorexia nervosa and a BMI of 14 kg/m² develops worsening postprandial epigastric pain and bilious vomiting causing rapid additional weight loss. Self-induced vomiting is initially suspected, but she denies purging behavior. Family and friends have witnessed the vomiting and support her history.

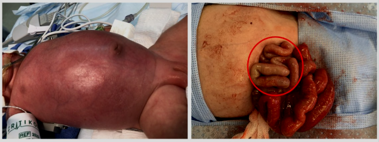
As her PCP, how would you proceed?

GO TO CONCLUSION ↓

Small Intestine Disorders (LO 1)

Necrotizing Enterocolitis (NEC)

NEC is the most common GI emergency in preterm neonates, especially those <32 weeks or <1500 grams. Pathogenesis is multifactorial: intestinal immaturity, microbial dysbiosis, formula feeding, and mucosal ischemia promote bacterial invasion and an exaggerated inflammatory response leading to bowel necrosis. Presentation typically occurs at 7–14 days of life with abdominal distension, feeding intolerance, and possibly bloody stools and clinical deterioration. Mortality may reach 50% in advanced disease. Breast milk feeding is the most effective preventive measure, reducing risk by ~50%, likely via microbiome effects. On examination, infants lack classic guarding or rebound. **Peritonitis instead is revealed as a distended, erythematous abdomen** (Figure 1). The pathognomonic finding on plain X-ray **is pneumatosis intestinalis (intramural gas)** (Figure 2).



Neonatal Peritonitis: red, distended abdomen
Inflamed small intestine with necrosis (circled)

Figure 1: NEC

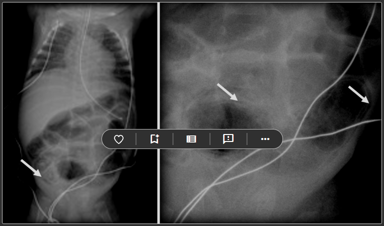


Figure 2. Pneumatosis Intestinalis on Abdominal X-ray. White arrows = air within the bowel wall.

Treatment: Most mild/early cases of NEC resolve with medical management alone — discontinue feeding , antibiotics, and supportive care. Surgery is reserved for perforation or clinical deterioration. Severe Necrotizing enterocolitis may result in necrotic bowel that requires surgical resection (Figure 1). Extensive intestinal loss can result in short bowel syndrome.

What is the most common abdominal examination finding in neonates with peritonitis?

Mesenteric Ischemia / Ischemic Small Intestine

Acute mesenteric ischemia (AMI) is a sudden reduction in blood flow to the intestines that can rapidly progress to bowel necrosis (Figure 3) with high mortality rates. There are four mechanisms:

- Arterial Embolism - the most acute, often from atrial fibrillation
- Arterial Thrombosis - chronic atherosclerotic disease of the SMA
- Mesenteric Venous Thrombosis - from hypercoagulable states
- Non-Occlusive Mesenteric Ischemia - NOMI, from low-flow states such as shock or vasopressors

Acute mesenteric ischemia classically presents with abdominal pain out of proportion to exam—the abdomen may initially be soft despite severe visceral pain. Early ischemia causes intense visceral pain from hypoperfusion while the exam remains benign because the parietal peritoneum is not yet involved. With progression to transmural necrosis, peritoneal irritation leads to peritonitis and sepsis. Elevated lactate is a late finding. CT angiography is the diagnostic test of choice (Figure 4). AMI primarily affects the SMA territory (jejunum and ileum), distinguishing it from ischemic colitis.

Treatment of acute mesenteric ischemia involves surgical embolectomy (or thrombectomy) to **restore blood flow**, followed by **resection of necrotic bowel** (Figure 3). Extensive resection carries a risk of **short bowel syndrome**.

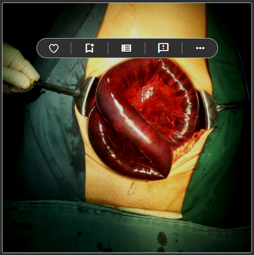


Figure 3. Necrotic small intestine

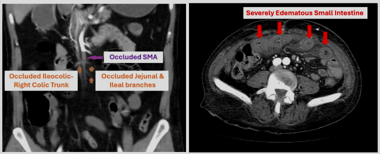

Occluded SMA
Occluded Ileocolic-Right Colic Trunk
Occluded Jejunal & Ileal branches
Severely Edematous Small Intestine

Figure 4. CT Angio reveals SMA occlusion and ischemic small intestine.

ACTIVE LEARNING QUESTION

An 83-year-old female with atrial fibrillation undergoes surgical exploration of the abdomen for peritonitis. The jejunum and ileum are ischemic. What is the most likely etiology?

A. ○ Atheroembolism from aorta to celiac artery

B. ○ Cardioembolic embolism to SMA

C. ○ Thrombosis of IMA

Explanation (requires correct answer)

Intussusception

Intussusception is telescoping of one bowel segment into another. In children (peak 5–9 months), most cases involve the ileocolic region and are typically idiopathic, often related to lymphoid hyperplasia acting as a lead point (Figure 5). The **classic triad—colicky abdominal pain, current jelly stool (from intestinal bleeding), and a palpable sausage-shaped mass**—is often incomplete. Bilious vomiting may occur due to small bowel obstruction. Lethargy or altered mental status can be an atypical but important presentation in infan **c target (donut) sign** (Figure 6). In adults, intussusception is uncommon and usually caused by a pathologic lead point such as a polyp or tumor.

CLINICAL CORRELATION

In Intussusception, a "lead point" is a mass that is pulled forward by peristalsis, causing one segment of bowel to telescope into another.

Treatment in children is pneumatic or hydrostatic enema reduction (air or saline under fluoroscopic or ultrasound guidance). Surgery is reserved for failed reduction, perforation, or suspected pathologic lead point.

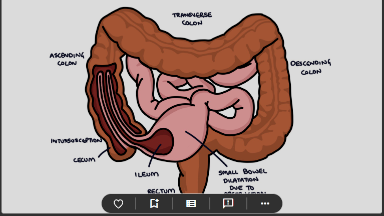


Figure 5. Intussusception

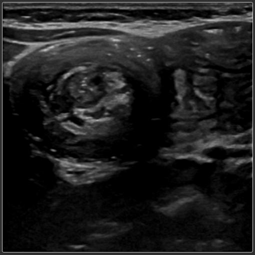
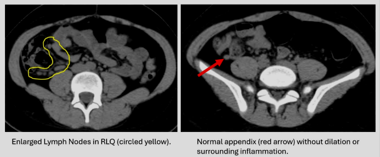


Figure 6. Ultrasound target sign of pediatric intussusception

Mesenteric Adenitis

Mesenteric adenitis is **inflammation of mesenteric lymph nodes**, usually triggered by a viral illness (often following an upper respiratory infection) or bacterial enteritis, classically *Yersinia enterocolitica*. It is a **common mimic of acute appendicitis** in children and is more frequent than appendicitis in the first decade, becoming uncommon after age 20. Presentation overlaps with appendicitis (right lower quadrant pain, fever, vomiting), but tenderness is less localized and may shift with repositioning. Peritoneal signs are less common, and labs often show lower WBC with relative lymphocytosis rather than neutrophilia. **Clinical differentiation is unreliable, and imaging is usually required.** Ultrasound shows enlarged mesenteric lymph nodes with a **normal appendix**. CT demonstrates clustered right lower quadrant nodes (≥5 mm) with a normal appendix (Figure 7). Mesenteric adenitis is self-limited, resolving within 1–4 weeks without any treatment.

Treatment is supportive with hydration and analgesics. Recognition is important to avoid unnecessary surgery.



Enlarged Lymph Nodes in RLQ (circled yellow).

Normal appendix (red arrow) without dilation or surrounding inflammation.

Figure 7. CT Mesenteric Adenitis and Normal Appendix

Appendicitis (L

Acute appendicitis is the most common cause of acute abdominal pain requiring surgery, with a lifetime risk of ~7–9% and peak incidence in the second and third decades. Pathophysiology begins with **luminal obstruction**—most often lymphoid hyperplasia in younger patients and **appendicoliths** in older patients; tumors may also cause obstruction in adults. Obstruction increases intraluminal pressure, compressing venous outflow and impairing capillary perfusion, leading to ischemia and bacterial invasion of the appendiceal wall (Figure 8). **If untreated, progression to gangrene and perforation occurs**, resulting in either a localized abscess (contained by the omentum) or generalized peritonitis.

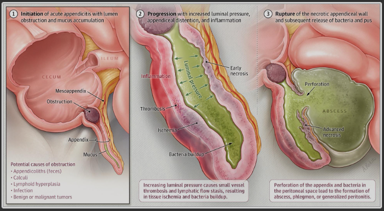

① **Initiation** of acute appendicitis with lumen obstruction and mucus accumulation
ILEUM
CECUM
Mesoappendix
Obstruction
Appendix
Mucus
Potential causes of obstruction
• Appendicoliths (feces)
• Calculi
• Lymphoid hyperplasia
• Infection
• Benign or malignant tumors
② **Progression** with increased luminal pressure, appendiceal distention, and inflammation
Luminal pressure
Early necrosis
Inflammation
Thrombosis
Ischemia
Bacteria buildup
Increasing luminal pressure causes small vessel thrombosis and lymphatic flow stasis, resulting in tissue ischemia and bacteria buildup.
③ **Rupture** of the necrotic appendiceal wall and subsequent release of bacteria and pus
Perforation
ABSCESS
Advanced necrosis
Perforation of the appendix and bacteria in the peritoneal space lead to the formation of abscess, phlegmon, or generalized peritonitis.

Figure 8. Pathophysiology of Acute Appendicitis

The classic presentation follows a predictable sequence. Pain **begins in the periumbilical region** because the distended appendix stimulates visceral afferent nerves traveling with the T8–T10 sympathetic fibers. Most patients then experience **migration of pain to the right lower quadrant** within 24 hours, as inflammation reaches the parietal peritoneum overlying McBurney's point (two-thirds the distance from the umbilicus to the anterior superior iliac spine). Anorexia is present in most (85%) patients and typically follows the onset of pain; nausea and vomiting may occur.

CLINICAL CORRELATION

Midgut visceral pain is referred to the periumbilical region because its visceral pain fibers enter the spinal cord at T10, which corresponds to the umbilical dermatome.

Physical exam signs include (Figure 9):

- **McBurney sign:** tenderness at the point 2/3 the way from the umbilicus to the anterior superior iliac spine - the most sensitive and specific sign
- **Rovsing sign:** RLQ pain with LLQ palpation - good specificity
- **Psoas sign:** RLQ pain with passive right hip extension, suggesting an inflamed retrocecal appendix
- **Obturator sign:** RLQ pain with internal rotation of the flexed right hip, suggesting an inflamed pelvic appendix.

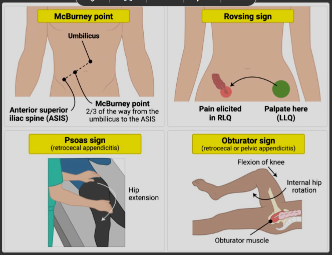

McBurney point
Umbilicus
Anterior superior iliac spine (ASIS)
McBurney point 2/3 of the way from the umbilicus to the ASIS
Rovsing sign
Pain elicited in RLQ
Palpate here (LLQ)
Psoas sign (retrocecal appendicitis)
Hip extension
Obturator sign (retrocecal or pelvic appendicitis)
Flexion of knee
Internal hip rotation
Obturator muscle

Figure 9. Examination of Appendicitis

Mild leukocytosis is present in most cases, typically with a left shift. Additional lab studies are used to rule out other etiologies, such as gallstone complications, urinary tract infection, or ectopic pregnancy.

Appendicitis remains a clinical diagnosis — history, exam, and laboratory findings are often sufficient, particularly in young males with classic presentations. However, **imaging has become ubiquitous** in practice and has reduced the negative appendectomy rate (proportion of patients who undergo appendectomy for suspected appendicitis but are found to have a normal appendix) from ~15% **the imaging procedure of choice is CT abdomen** (Figure 10). **Ultrasound is preferred in children and pregnant patients to avoid radiation** (Figure 11).

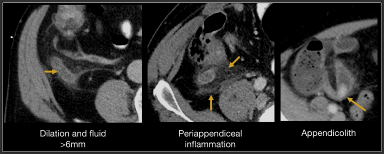


Figure 10. CT findings of appendicitis

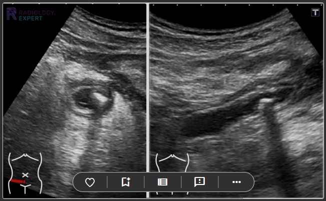


Figure 11. Ultrasound of RLQ revealing enlarged appendix in short and long axis, thickened appendix wall, and appendicolith with acoustic shadowing

Appendicitis is classified as uncomplicated (no perforation, abscess, or phlegmon) or complicated (perforation with abscess or phlegmon). Symptoms lasting >24 hours increase perforation risk, although perforation can occur earlier.

Treatment: Laparoscopic appendectomy is the definitive treatment, offering faster recovery and lower surgical site infection than open surgery (Figure 12). Preoperative antibiotics covering gram-negative rods and anaerobes (e.g., single-dose cefoxitin or cefotetan, or cefazolin plus metronidazole) are standard. Postoperative antibiotics are unnecessary unless perforation has occurred, which increases the risk of postoperative abscess.

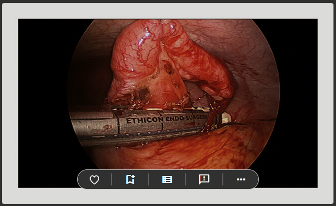


Figure 12. Laparoscopic Appendectomy

CLINICAL REASONING

An antibiotics-first (non-surgical) approach is a viable alternative for selected patients with uncomplicated appendicitis. Outcomes are comparable to appendectomy, though ~30% of patients have recurrence within 90 days requiring surgery. Guidelines support offering antibiotics to patients without high-risk CT findings (appendicolith, mass, dilated appendix) while recommending surgery for those with high-risk features.

In addition to gastroenteritis, what is the other leading cause of RLQ pain in a child with a normal appendix?

LARGE INTESTINE TOPICS

Colon Disorders (LO 3)

Ischemic Colitis

Ischemic colitis is more common and less life-threatening than mesenteric ischemia and predominantly affects the elderly. It results from transient reductions in colonic blood flow due to hypoperfusion, vasospasm, or small-vessel occlusion, **most often involving watershed areas (splenic flexure and rectosigmoid junction).** Presentation is acute **abdominal pain followed**

by hematochezia or bloody diarrhea within 24 hours. CT shows colonic wall thickening with pericolonic fat stranding. Most cases (~85%) are self-limited and resolve with supportive care. Colonoscopy may help assess severity, showing dusky, ulcerated, or gangrenous mucosa (Figure 13). Gangrenous disease requires emergent surgical resection due to the risk of perforation, sepsis, and death.

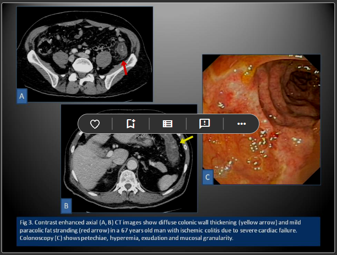


Fig 3. Contrast enhanced axial (A, B) CT images show diffuse colonic wall thickening (yellow arrow) and mild paracolic fat stranding (red arrow) in a 67 years old man with ischemic colitis due to severe cardiac failure. Colonoscopy (C) shows petechiae, hyperemia, exudation and mucosal granularity.

Figure 13. Ischemic colitis, likely a self-limited case

Diverticulosis, Diverticulitis

Diverticulosis refers to colonic outpouchings - **pseudodiverticula**, involving only mucosa and submucosa - which herniate through the muscular layer at **sites of vasa recta penetration near the tenia coli** (Figure 14). The rectum, which has a continuous outer longitudinal muscle layer, does not develop diverticula (Figure 15). Diverticulitis is thought to result from chronically elevated intraluminal pressure particularly in the sigmoid colon, likely related to segmentation of firm stools associated with low-fiber diets. In the US, diverticulosis is common with advancing age and is largely asymptomatic; only ~4–5% of patients develop diverticulitis.

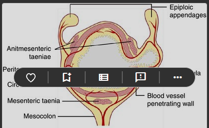
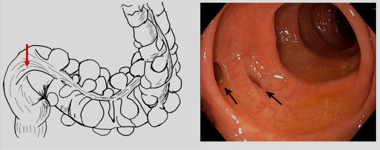


Figure 14. Colonic diverticula develop at the sites where penetrating arteries traverse the colon wall.



Left: The sigmoid colon transitions to the rectum where the tenia converge to create a complete outer longitudinal muscle layer (red arrow). Without tenia, the rectum is not prone to diverticulosis. Right: Diverticula are commonly seen on colonoscopy (black arrows) and the vast majority of patients never develop diverticulitis.

Figure 15. Diverticulosis

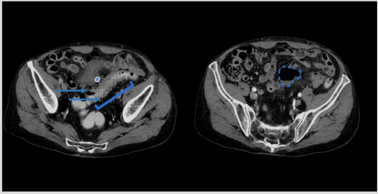
CLINICAL REASONING

The belief that nuts, seeds, and popcorn cause diverticulitis is a **myth** — large prospective studies show no increased risk, and some studies show decreased risk with popcorn int

No popcorn kernel or seed has ever been seen in an inflamed diverticulum.

Diverticulitis is inflammation and microperforation of a diverticulum. In Western populations, it is predominantly left-sided (sigmoid colon), while in Asian populations, right-sided disease is more common. The classic presentation is left lower quadrant pain with fever, leukocytosis, and LLQ tenderness. **CT with IV contrast is the gold standard for diagnosis** (Figure

16). Diverticulitis is classified as uncomplicated (~85% of cases, localized inflammation only) or complicated (~12–15%, with abscess, perforation, fistula, obstruction, or stricture).



CT Abdomen Pelvis with contrast: diverticulitis in a segment of sigmoid colon (bracket), with extensive diverticulosis (arrows), wall thickening (asterisk), and a nearby abscess containing mostly air (dotted circle).

Diverticular bleeding is a separate entity — painless, brisk hematochezia from arterial rupture at the neck of a diverticulum. Rectal bleeding is not a feature of acute diverticulitis. Like most sources of lower GI bleeding, it is self-limited in the overwhelming majority of cases.

CLINICAL REASONING

LLQ pain, tenderness, *and rectal bleeding* is consistent with ischemic colitis, not diverticulitis.

Uncomplicated diverticulitis is treated with antibiotics (for Gram-negative rods and anaerobes); complicated cases (abscess, perforation, peritonitis) require drainage or surgery. Patients with multiple prior episodes have a high likelihood of additional episodes. No pharmacologic or dietary therapy has proven effective for preventing recurrence. Elective sigmoid colectomy may be offered based on episode frequency, severity, and quality-of-life impact - it is the patient's decision whether to proceed with surgical treatment.

CLINICAL CORRELATION

The sigmoid colon, [illegible]ere the tenia converge to create a complete outer longitudinal muscle layer. Without tenia, the rectum is not prone to diverticulosis.

When sigmoid colectomy is performed to minimize the recurrence risk of diverticulitis, the resection must be into the proximal rectum to achieve a very low recurrence rate — recurrence after colectomy is most commonly due to inadequate distal resection of the high-pressure sigmoid colon.

Angiodysplasia

Angiodysplasias (angioectasias) are dilated, thin-walled mucosal and submucosal blood vessels, most commonly in the cecum and right colon. They are acquired lesions seen predominantly in elderly patients and also occur in the small intestine, where they are the most common cause of obscure GI bleeding. Presentation is **painless bleeding—either occult blood loss with iron deficiency anemia** or **overt hematochezia**. Bleeding stops spontaneously in most cases (~85%) but often recurs. When treatment is required, endoscopic thermal coagulation (e.g., argon plasma coagulation) is first-line (Figure 17).

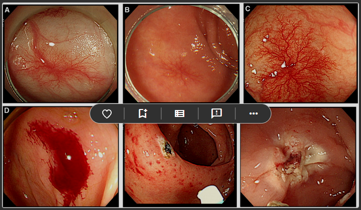


Figure 17. Colonic Angiodysplasias: A, B, C without bleeding. D bleeding. E, F, treated during colonoscopy

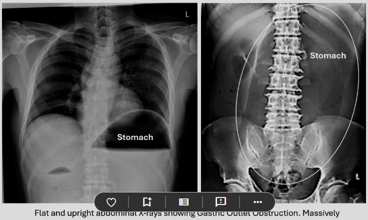
An older adult presents with acute hematochezia and LLQ pain. What is the most likely etiology?

Gastrointestinal Obstructions (LO 4)

Gastric Outlet Obstruction (GOO)

Gastric outlet obstruction (GOO) is a mechanical blockage at the pylorus or proximal duodenum. [illegible] (obstruction proximal to the ampulla of Vater), epigastric distension, early satiety, and weight loss. **Prolonged vomiting causes hypochloremic, hypokalemic metabolic alkalosis.** In adults, the most common causes are peptic ulcer disease and gastric or pancreatic malignancy. In infants, the classic cause is pyloric stenosis—hypertrophy of the pyloric muscle presenting at 3–6 weeks with progressive, projectile, nonbilious vomiting. Plain films reveal a markedly distended stomach with minimal distal gas **(Figure 18)**, and an upper GI can confirm the level of obstruction **(Figure 19)**. In infants, ultrasound confirms pyloric stenosis (thickened pyloric muscle). In adults, CT identifies the obstruction and is better at identifying the cause **(Figure 20)**. Upper endoscopy and biopsy will definitively confirm the diagnosis.

Treatment of GOO addresses the underlying cause — pyloromyotomy for pyloric stenosis, surgery for PUD or malignancy.



Flat and upright abdominal X-rays showing Gastric Outlet Obstruction. Massively distended stomach with air fluid level on upright film. Minimal distal gas.

Figure 18. GOO on plain films

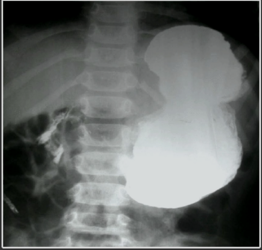


Figure 19. GOO on Upper GI. Contrast does not leave the distended stomach.

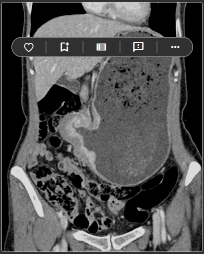


Figure 20. CT Abdomen with GOO due to distal gastric cancer

Superior Mesenteric Artery (SMA) Syndrome

SMA syndrome is compression of the third portion of the duodenum between the superior mesenteric artery and aorta due to narrowing of the aortomesenteric angle (<25°) from loss of the intervening fat pad (Figure 21). This is typically caused by rapid weight loss (e.g., eating disorders, catabolic illness, burns). Patients present with postprandial epigastric pain, nausea, bilious vomiting, early satiety, and weight loss. Symptoms worsen when supine and improve in the prone or left lateral decubitus position. Upper GI series shows abrupt duodenal cutoff with proximal dilation. CT confirms the diagnosis by demonstrating a reduced aortomesenteric angle and duodenal compression (Figure 22).

Treatment is improved nutrition to restore retroperitoneal fat and widen the angle. Leaning forward in a knee-to-chest position after eating will help. If conservative management fails, surgical treatment is offered to bypass the obstruction.

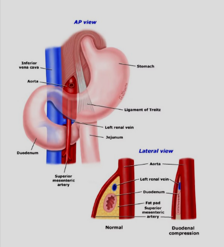
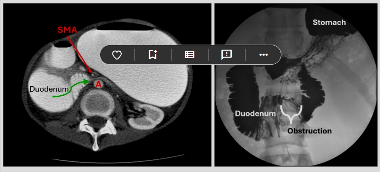


Figure 21. SMA Syndrome



CT Abdomen with IV and po contrast: The 3rd portion of the duodenum is pinched between the Aorta and SMA.

Upper GI: The 3rd portion of the duodenum is partially obstructed as it crosses the midline.

Figure 22. SMA Syndrome on CT Abdomen and Upper GI

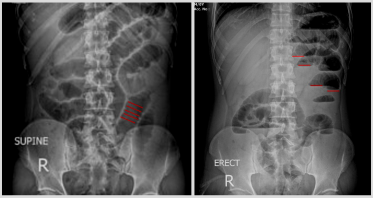
Small Bowel Obstruction (SBO)

Small bowel obstruction (SBO) accounts for most mechanical bowel obstructions. Obstruction leads to proximal accumulation of gas and fluid with bowel dilation. Strangulation—vascular compromise from the obstructing lesion (e.g., adhesive band, hernia)—can cause ischemia even before significant dilation develops. The classic presentation is colicky abdominal pain, nausea, bilious vomiting, abdominal distension, and obstipation. Bowel sounds may be high-pitched and “tinkling.” The abdomen is distended and tympanitic to percussion. Mild tenderness may be present, but peritoneal signs (rigidity) should be absent in uncomplicated cases; their presence suggests strangulation or perforation and requires urgent surgical evaluation.

Small Bowel Obstruction is suspected clinically and confirmed by imaging. Plain films can show **dilated air-filled small bowel loops, air-fluid levels, and a paucity of colonic gas** (Figure 23), but **CT abdomen** has become the test of choice, offerin ause, and severity of obstruction — including detection of strangulation (compromised blood supply) (Figure 24).

CLINICAL CORRELATION

On plain abdominal X-ray, the small intestine is identified by the presence of its **closely spaced circular folds that traverse the width of the intestine.** An SBO will show distended small intestine, air-fluid levels with stair-stepping on the upright film, and a lack of gas in the colon.



Flat and Upright Abdominal X-rays revealing SBO: distended small intestine (circular folds are close together and usually traverse the bowel width). Upright film shows air-fluid levels with stair-stepping. There is no gas entering the colon – all the visualized bowel is small intestine.

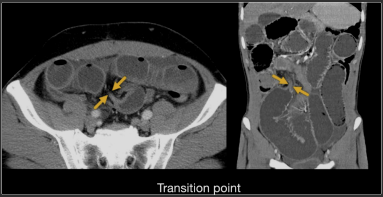
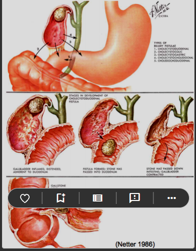


Figure 24. CT Abdomen/Pelvis revealing SBO

The most common causes of SBO:

- **Adhesions** are the most common cause (60–75%), forming after abdominal or pelvic surgery any time in the past (even decades ago).
- **Hernias** with intestinal incarceration are the second most common cause—inguinal, femoral, umbilical, incisional, or internal.
  - **Hernia contents must be reduced** - gently pushed back into the abdomen. If a hernia contains incarcerated (unable to be reduced) intestine, surgical repair is emergent due to the risks of intestinal strangulation (loss of blood supply).
- **Neoplasms** are r y can obstruct by luminal narrowing, intussusception lead points, or peritoneal carcinomatosis.
- **Crohn disease** (~5%) causes SBO through acute inflammatory edema or chronic fibrotic strictures.
- **Gallstone ileus** is a rare cause of SBO, yet it is a classic topic for board exams. In this disorder, a large gallstone erodes through the gallbladder wall into the duodenum and creates a cholecystoenteric fistula. The gallstone is propelled distally in the small intestine. If it is large, it impacts at the ileocecal valve obstructing the small intestine (Figure 25). The Rigler **triad on imaging includes pneumobilia** (air in the biliary tree because the biliary system has a large, open connection to the GI tract), **SBO, and an ectopic gallstone** in the intestine (Figure 26).
- Other causes include intussusception with a pathologic lead point and radiation enteritis.


F. Netter M.D.
©CIBA
TYPES OF BILIARY FISTULAE
1. CHOLECYSTODUODENAL
2. CHOLECYSTOCOLIC
3. CHOLECYSTOGASTRIC
4. CHOLECYSTOCHOLEDOCHAL
5. CHOLEDOCHODUODENAL
1
2
3
4
5
STAGES IN DEVELOPMENT OF CHOLECYSTODUODENAL FISTULA
GALLBLADDER INFLAMED, DISTENDED, ADHERENT TO DUODENUM
FISTULA FORMED; STONE HAS PASSED INTO DUODENUM
STONE HAS PASSED DOWN INTESTINE; GALLBLADDER CONTRACTED
GALLSTONE

(Netter 1986)

Figure 25. Gallstone Ileus

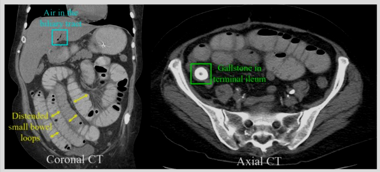


Figure 26. CT Abdomen/Pelvis with gallstone obstructing distal ileum. Pneumobilia is noted.

Treatment: Initial management of adhesive SBO is **nonoperative: NPO, nasogastric decompression, and IV fluid resuscitation**; approximately **85% will resolve** without surgery. **Surgical intervention** is indicated for failure to resolve, signs of strangulation or peritonitis, or unknown etiology (e.g., no prior surgical history to explain adhesions).

What is the most common cause of bilious vomiting and abdominal distention in an adult with imaging showing dilated small intestine and no gas in the colon?

Colonic Obstruction

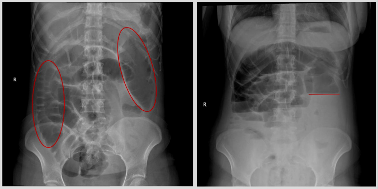
Colonic obstruction accounts for ~25% of mechanical bowel obstructions. If the ileocecal valve is competent, a closed-loop obstruction occurs and the colon cannot decompress into the small intestine. By Laplace's law, the cecum (largest diameter) develops the highest wall tension and is most prone to perforation; a cecal diameter >9 cm is concerning. Compared with small bowel obstruction, colonic obstruction presents more insidiously, with early abdominal distension and obstipation, while vomiting is a late finding.

The main causes of colon obstruction include:

- **Colorectal cancer** is the most common cause, particularly in the distal colon where the smaller lumen causes earlier obstruction.
- **Volvulus** is the most common benign cause; sigmoid volvulus predominates and is discussed separately below.
- **Diverticular stri** e — chronic fibrosis from recurrent diverticulitis progressively narrows the sigmoid lumen and can mimic malignancy on imaging.

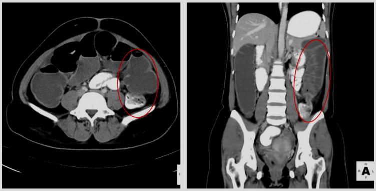
Diagnosis: Plain abdominal films show **dilated colon proximal to the obstruction with absent distal gas** (Figure 27). **CT abdomen** is the test of choice — it identifies the site, cause, and severity of obstruction and can detect complications such as ischemia or perforation (Figure 28). Colonoscopy can be performed, advancing the scope to the level of obstruction to confirm the etiology.

Treatment depends on the etiology, with colon cancer, cecal volvulus, and benign strictures requiring surgical resection. Sigmoid volvulus can be decompressed by colonoscopy.



Flat and Upright Abdominal X-rays revealing colon obstruction: distended colon has haustral markings about 1 cm apart, they do not traverse the width of the bowel (red circles). The upright film shows an air-fluid level at above the Obstruction (red line). Neither film shows any distal colon gas beyond the mid descending colon. Scattered small bowel gas Is also noted, cons [illegible]

Figure 27. Plain films revealing colonic obstruction

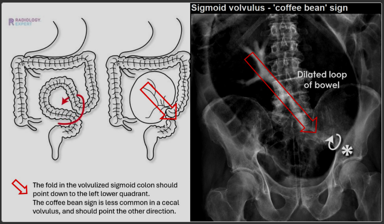


CT Abdomen/Pelvis with IV, oral, and rectal contrast: Distended colon is noted in both the axial and coronal views, with a tight narrowing (an adenocarcinoma) in the descending colon (red circles). Rectal contrast (white) does not pass proximally through the narrowing.

Figure 28. CT Abdomen/Pelvis revealing colon obstruction

Colon Volvulus

Volvulus is twisting of the bowel around its mesenteric axis, causing a closed-loop obstruction with risk of ischemia. Sigmoid volvulus is the most common type of volvulus in the US and is associated with elderly, institutionalized patients with chronic constipation and a redundant sigmoid colon. The classic radiographic finding is the “coffee bean sign” on abdominal X-ray (Figure 29). Abdomen CT is more accurate than plain film. Initial treatment is endoscopic decompression, followed by elective sigmoid resection to prevent recurrence. Cecal volvulus occurs in younger patients and requires surgical intervention, as endoscopic decompression is ineffective.



The fold in the volvulized sigmoid colon should point down to the left lower quadrant.
The coffee bean sign is less common in a cecal volvulus, and should point the other direction.

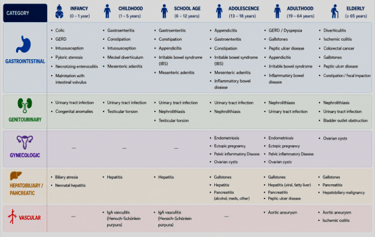
Figure 29. X-ray of sigmoid volvulus

CASE CONNECTION BACK TO INTRODUCTION ↑

You obtain a CT abdomen/pelvis with oral contrast. A markedly distended stomach and duodenum are seen with compression of the third portion of the duodenum between the aorta and the superior mesenteric artery.

You diagnose SMA syndrome and recommend inpatient nutritional rehabilitation, leaning forward afte [illegible] a 6-kg weight gain, her vomiting resolves.

Consider the various causes of abdominal pain and how they relate to patient age (Figure 30):



COMMON CAUSES OF ABDOMINAL PAIN BY AGE

CATEGORY	INFANCY (0 – 1 year)	CHILDHOOD (1 – 5 years)	SCHOOL AGE (6 – 12 years)	ADOLESCENCE (13 – 18 years)	ADULTHOOD (19 – 64 years)	ELDERLY (≥ 65 years)
GASTROINTESTINAL	• Colic • GERD • Intussusception • Pyloric stenosis • Necrotizing enterocolitis • Malrotation with intestinal volvulus	• Gastroenteritis • Constipation • Intussusception • Meckel diverticulum • Mesenteric adenitis	• Gastroenteritis • Constipation • Appendicitis • Irritable bowel syndrome (IBS) • Mesenteric adenitis	• Appendicitis • Gastroenteritis • Constipation • Irritable bowel syndrome (IBS) • Mesenteric adenitis • Inflammatory bowel disease	• GERD / Dyspepsia • Gallstones • Peptic ulcer disease • Appendicitis • Irritable bowel syndrome • Inflammatory bowel disease	• Diverticulitis • Ischemic colitis • Colorectal cancer • Gallstones • Peptic ulcer disease • Constipation / Fecal impaction
GENITOURINARY	• Urinary tract infection • Congenital anomalies	• Urinary tract infection • Testicular torsion	• Urinary tract infection • Nephrolithiasis • Testicular torsion	• Urinary tract infection • Nephrolithiasis	• Nephrolithiasis • Urinary tract infection	• Nephrolithiasis • Urinary tract infection • Bladder outlet obstruction
GYNECOLOGIC	—	—	—	• Endometriosis • Ectopic pregnancy • Pelvic inflammatory Disease • Ovarian cysts	• Endometriosis • Ectopic pregnancy • Pelvic inflammatory Disease • Ovarian cysts	• Ovarian cysts
HEPATOBILIARY / PANCREATIC	• Biliary atresia • Neonatal hepatitis	• Hepatitis	• Hepatitis	• Gallstones • Hepatitis • Pancreatitis (alcohol, meds, other)	• Gallstones • Hepatitis (viral, fatty liver) • Pancreatitis • Peptic ulcer disease	• Gallstones • Pancreatitis • Hepatobiliary malignancy
VASCULAR	—	• IgA vasculitis (Henoch-Schönlein purpura)	• IgA vasculitis (Henoch-Schönlein purpura)	—	• Aortic aneurysm	• Aortic aneurysm • Ischemic colitis

Figure 30.

Summary

- Acute mesenteric [illegible]th high mortality, classically presenting with pain out of proportion to exam; CT angiography is the diagnostic modality of choice.
- Necrotizing enterocolitis (NEC) is the most common GI emergency in preterm neonates, with pneumatosis intestinalis as the pathognomonic radiographic finding.
- Angiodysplasia is the most common vascular lesion of the GI tract and the leading cause of obscure small bowel bleeding in older adults.
- Acute appendicitis is the most common cause of acute abdominal pain requiring surgery; laparoscopic appendectomy is the definitive treatment.
- Ischemic colitis is the most common form of intestinal ischemia, preferentially affecting watershed areas of the colon and typically presenting with self-limited acute abdominal pain followed by hematochezia.
- Bowel obstruction is classified as small bowel (75%, most commonly from adhesions) or colonic (25%, most commonly from colorectal cancer); key emergencies include strangulation, cecal perforation in closed-loop obstruction, and volvulus, with sigmoid volvulus treated initially by endoscopic decompression and cecal volvulus requiring surgical intervention.

Review Questions

1. **A premature 10-day-old NICU infant develops abdominal distention, bloody stools, metabolic acidosis, and shock. Abdominal x-ray demonstrates pneumatosis intestinalis. Despite bowel rest and IV antibiotics, the infant develops worsening abdominal erythema and pneumoperitoneum. Which of the following is [illegible]**

   A. ○ Congenital absence of intestinal ganglia

   B. ○ Hypertrophy of the pyloric muscularis propria

   C. ○ Intestinal ischemia with bacterial invasion of the bowel wall

   D. ○ Telescoping of the terminal ileum into the colon

   Explanation (requires correct answer)

2. **A 19-year-old man develops vague abdominal pain that several hours later localizes to the right lower quadrant with nausea and fever. CT abdomen/pelvis reveals a 9 mm tubular structure in the RLQ with surrounding inflammation. In this condition, the visceral pain is referred to which dermatome level?**

   A. ○ T8

   B. ○ T10

   C. ○ T12

   D. ○ L2

   Explanation (requires correct answer)

3. **A 75-year-old man has had repeated episodes of LLQ pain and tenderness. He has not experienced bleeding. Each episode resolves with antibiotic treatment directed at gram-negative rods and anaerobes. Which pathology**

has led to these episodes?

- **A.** ○ Colonic mucosal outpouchings occurring at sites where penetrating vasa recta weaken the bowel wall
- **B.** ○ Dilated fragile submucosal blood vessels causing intermittent lower GI bleeding in older adults
- **C.** ○ Transient hypoperfusion injury of watershed regions of the colon
- **D.** ○ Twisting of the sigmoid colon around its mesentery causing large bowel obstruction and ischemia

Explanation (requires correct answer)

4. A 9-year-old boy presents to the ED with colicky abdominal pain, bilious vomiting, and abdominal distention. Abdominal X-rays reveal distended small intestine with air-fluid levels and stair-stepping; no distal gas is seen. Which additional physical examination finding would most strongly suggest a cause of obstruction amenable to immediate bedside intervention?

- **A.** ○ Abdominal rigidity
- **B.** ○ Bulge in groin
- **C.** ○ Scar on abdomen
- **D.** ○ Tinkling bowel sounds

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- **01** ☐ **Differentiate small intestine disorders by presentation, pathophysiology, diagnosis, and treatment.**
- **02** ☐ **Analyze the presentation, pathophysiology, diagnosis, and treatment of appendicitis.**
- **03** ☐ **Differentiate colon disorders by presentation, pathophysiology, diagnosis, and treatment.**
- **04** ☐ **Differentiate gastrointestinal obstructions by presentation, pathophysiology, diagnosis, and treatment.**

Explanation (requires correct answer)

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Objective Confidence: 0/4